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NormAE: Deep Adversarial Learning Model to Remove Batch Effects in Liquid Chromatography Mass Spectrometry-Based Metabolomics Data

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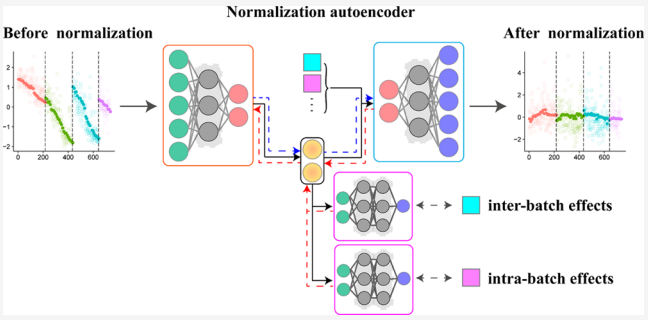
ABSTRACT: Untargeted metabolomics based on liquid chromatography−mass spectrometry is affected by nonlinear batch effects, which cover up biological effects, result in nonreproducibility, and are difficult to be calibrate. In this study, we propose a novel deep learning model, called Normalization Autoencoder (NormAE), which is based on nonlinear autoencoders (AEs) and adversarial learning. An additional classifier and ranker are trained to provide adversarial regularization during the training of the AE model, latent representations are extracted by the encoder, and then the decoder reconstructs the data without batch effects. The NormAE method was tested on two real metabolomics data sets. After calibration by NormAE, the quality control samples (QCs) for both data sets gathered most closely in a PCA score plot (average distances decreased from 56.550 and 52.476 to 7.383 and 14.075, respectively) and obtained the highest average correlation coefficients (from 0.873 and 0.907 to 0.997 for both). Additionally, NormAE significantly improved biomarker discovery (median number of differential peaks increased from 322 and 466 to 1140 and 1622, respectively). NormAE was compared with four commonly used batch effect removal methods. The results demonstrated that using NormAE produces the best calibration results.



Metabolomics is an omics technology that uses state-of-the-art analytical instrumentation in conjunction with pattern recognition techniques to monitor and discover metabolic changes in subjects related to disease status or in response to a medical or external intervention.[1] It is a powerful tool for understanding changes in metabolism, and can also lead to an improved understanding of the gene and protein pathways that regulate metabolite levels.[2] One of the strategies developed to achieve comprehensive coverage in metabolomics study is untargeted metabolomics, which is driven by an unbiased approach in which as many metabolites as possible are measured and compared between samples.[3] Because untargeted metabolomics provides global information about metabolites in biological samples, it is important for biological-related studies, for example, the exploration of functional compounds and discovery of biomarkers.[4,5] At present, liquid chromatography−mass spectrometry (LC-MS) has become the most commonly applied tool for untargeted metabolomics because it can be used to achieve rich metabolome information with minimum sample pretreatment required.[4]

However, untargeted metabolomics studies using LC-MS measurements are affected by batch effects. Batch effects are defined as systematic technical differences that occur when samples are processed and measured in different batches and are unrelated to any biological variation.[6] Intrabatch effects[7] in metabolomics studies, which are defined as a class of batch effects that gradually change the instrumental response during the measurement of a batch of samples, are frequently observed as a result of a buildup of contaminants. Additionally, interbatch effects can occur when a large-scale study has to be performed in batches, and result from column replacement and instrument calibration in LC-MS experiments. These systematic effects are nonlinear and become entangled with real biological effects, which results in a reduction of statistical power and concomitant nonreproducibility. These unwanted systematic variations are a critical obstacle for untargeted LC-MS metabolomics study at the present time.

There are several methods that aim to remove batch effects in untargeted metabolomics data. A common approach is to use repeated assay of quality control samples (QCs) from a

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consistent control pool.[7,8] QCs are pooled samples composed of small aliquots of each sample from biological subjects,[9] which are supposedly representative of the subject samples under study. QCs should have strong correlation relationships with each other because they are essentially identical. They are inserted regularly or randomly within batches and monitor the stability of the metabolomics system. Then models that describe the relationships between intensities and injection orders in QCs are fitted and applied in biological samples to remove batch effects. QC-RLSC[8] and QC-SVRC[7] are the two most common QCs-based methods. The former builds a nonlinear robust smoothing spline function to predict the batch effects of each peak, whereas the latter fits support vector machine regression. Recently, SERRF,[10] another QCs-based method that uses the Random Forest (RF), was proposed and outperformed other existing methods. However, QCs-based methods have the limitation of poor generalization in subject samples of fitted models that use QCs. Overfitting results from the small number of QCs for the training models and cannot be avoided. Therefore, instead of using QCs, well-known methods are introduced from other omics areas, particularly genomics, that can remove batch effects from subject samples directly.[11] They can be classified into two main approaches: location-scale methods and matrix factorization methods. Location-scale methods assume a model for data distribution within a batch, and adjust the data within each batch to fit this model. This approach is the most straightforward, and many methods have been proposed, such as ratio-based methods,[12] ComBat,[13] quantile based methods,[14] and mean or median centering.[15] Matrix factorization methods, such as SVD,[16] ICA,[17] and EigenMS,[18] assume that the metabolites-by-samples matrix can be represented by a small set of one-rank components that can be estimated using matrix factorization. The components that correlate with the batch labels are then removed to obtain a data set without batch effects. These methods from other omics areas are dependent on some assumptions, such as there are no intrabatch effect and the model is linear, that can be easily violated for metabolomics data. It is not suitable for them to be directly applied to metabolomics data.[19] To overcome this problem, WaveICA,[19] which is a new matrix factorization method that first transforms data using a wavelet transform (WT) and then uses ICA to remove batch effects, was proposed. The application of a WT results in nonlinear analytical capabilities and improves performance. However, only effects related to frequency can be detected by a WT, other batch effects are still hidden in the data. Therefore, a method with a stronger nonlinear fitting ability is required that uses as many samples as possible.

Recently, deep neural networks (DNN) have made breakthroughs in various scientific areas such as image classification,[20] text recognition,[21] and speech recognition.[22] DNNs consist of successive elementary operations, called layers, which compute increasingly more complex features by taking the results of preceding operations as input.[23] Highly nonlinear and complex features can be modeled by DNNs with many hidden layers. DNNs have been used in several bioinformatics applications, such as predicting protein binding sites in DNA and RNA,[24] and single-cell DNA methylation states.[25] There has been research in which the batch effect of single-cell RNA-seq data[26] was removed using deep generative models;[27] however, no related work has been published for metabolomics.

In this study, we propose a novel deep learning model called the normalized autoencoder (NormAE) to remove batch effects of untargeted metabolomics data. It is inspired by the autoencoder (AE) and adversarial learning[23] from deep learning, and can remove nonlinear batch effects without overfitting. We used two LC-MS untargeted metabolomics data sets to evaluate our method: one contained a total of 644 plasma samples in four batches with 85 QCs and the other contained 644 plasma samples in four batches with 81 QCs. The numbers of detected peaks were 8113 and 8194, respectively. We also compared the proposed method with four common batch effect removal methods: QC-RLSC, ICA, ComBat, and WaveICA. The results demonstrated that NormAE can remove more batch effects nonlinearly and screen for more potential metabolic peaks than the other methods.

■ EXPERIMENTAL SECTION

Plasma Collection. A total of 644 patients who gave informed consent were recruited from the Department of Colorectal Surgery of the Affiliated Tumor Hospital at Harbin Medical University between May 2015 and May 2016. They consisted of 73 chronic enteritis (CE) patients and 571 colorectal cancer (CRC) patients who were adenocarcinoma patients. None of the participants had other metabolic diseases or inflammatory conditions at the beginning of study. The study was approved by the Ethics Committee of the Affiliated Tumor Hospital of Harbin Medical University (Harbin, China). A volume of 5 mL of whole blood was collected from each participant into an EDTA tube after fasting for 12 h. The supernatant was extracted by centrifuging fresh blood at 1323 g for 10 min, 4 °C, and then stored at −80 °C until ready to proceed.

Sample Preparation and Pretreatment. Before preparation and UHPLC-QTOF/MS analysis, all plasma samples were randomized. QCs were generated by mixing small aliquots of all plasma samples. Blank samples (75% MeOH in water) and QCs were injected into every eight samples during acquisition.

A volume of 50 μL of each plasma sample was placed in an individual well in a 96-well plate after the plasma samples were thawed at 4 °C on ice. The plasma was then extracted with 150 μL of MeOH (which was kept at −20 °C before extraction) using the Bravo liquid handling system (Agilent Technologies, Santa Clara, CA), followed by vortexing for 30 s and incubation for 2 h at −20 °C to precipitate proteins. The 96-well plate was then sent to the centrifugal machine to centrifuge at 4000 rpm for 20 min at 4 °C. The resulting supernatants were transferred to LC-MS vials and stored at −80 °C for further UHPLC-QTOF/MS analysis.

Chromatography and Mass Spectrometry. The analysis of all plasma samples was performed using a UHPLC system (1290 series, Agilent Technologies) coupled to a quadrupole time-of-flight (QTOF) mass spectrometer (Agilent 6550 iFunnel Q-TOF, Agilent Technologies). Waters Acquity UHPLC HSS T3 columns and Agilent Amide columns were used in our analysis for LC separation, and the temperature of the columns was maintained at 25 °C during the entire process. The flow rate of the mobile phase was 0.5 mL/min and the sample injection volume was 1 μL. The mobile phase A was 0.1% FA in water in T3 positive mode or 25 mM NH_4OH and 25 mM NH_4OAc in water in Amide positive

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mode, and B was 0.1% FA in CAN in T3 positive mode or 100% CAN in Amide positive mode.

Data Preprocessing and Annotation. Raw MS data (.d) files were first converted to mzXML format using ProteoWizard. The R package XCMS (version 3.2) was then used to preprocess these files with the mzXML format and generate a data matrix that consisted of the retention time, mass-to-charge ratio (m/z) values, and peak intensity. After XCMS data processing, the R package CAMERA was used to annotate the peaks. We discarded metabolic features that were detected in less than 80% of the subject samples to prevent them from influencing the results.

Data Description. After the above data processing, 8113 peaks in Amide mode and 8194 in T3 mode were present in the final data sets. A total of 644 subject samples were separated into four batches that consisted of 192, 192, 184, and 76 samples. Additionally, there were 25, 25, 24, and 11 QCs in the Amide data set and 25, 25, 21, and 10 QCs in the T3 data set, respectively.

The Amide column had poorer stability than that of T3, so there were an extra few QCs in the Amide data set. However, more extremely polar species were retained by the Amide column. The two data sets were slightly different in terms of the metabolite properties and stability. Using both data sets ensured the rationality and universality of the comparison.

■ WORKFLOW AND METHODS

Deep Neural Networks. DNNs are machine learning models that consist of successive hidden layers, such as multiple linear models with their output transformed by a nonlinear activation function, which compute increasingly more complex features by taking the results of preceding operations as input.[23] Together, these layers compose the input features into relevant complex patterns, which facilitates tasks, such as distinguishing two classes (Figure 1A). DNNs can be typically trained using backpropagation algorithms,[28] which are a family of methods from stochastic gradient descent that are suited to training multilayer models on large data sets.

Autoencoder. AEs[29] are a family of DNNs for unsupervised learning, which learn latent representations of data by minimizing the reconstruction loss between the input and output. AEs mainly consist of two parts, an encoder and decoder, which are both multilayer nonlinear neural networks (Figure 1B). The encoder embeds original high-dimensional data into low-dimensional space, which is called the bottleneck layer,[23] and then the decoder transforms the values of the bottleneck layer back to the original high-dimensional space. The parameters of AEs are trained by minimizing the differences between the original data and the reconstruction, which is quantified by a loss function, such as the mean square error (MSE), and mean absolute error (MAE). This approach forces the network to extract useful features of data, as the bottleneck layer with low dimensions makes it infeasible to learn the perfect reconstruction.[23] Reconstructing the data is often interpreted as denoising because unimportant variations are automatically omitted.

Normalization Autoencoder (NormAE). We use a nonlinear AE to remove batch effects. Inspired by generative adversarial networks (GAN),[27] this task can be completed via adversarial training with additional discriminators.

Suppose there are batch labels Y^b and interbatch effects are characterized by them. A special AE with encoder E and decoder D is defined (Figure 1C). E can encode the original

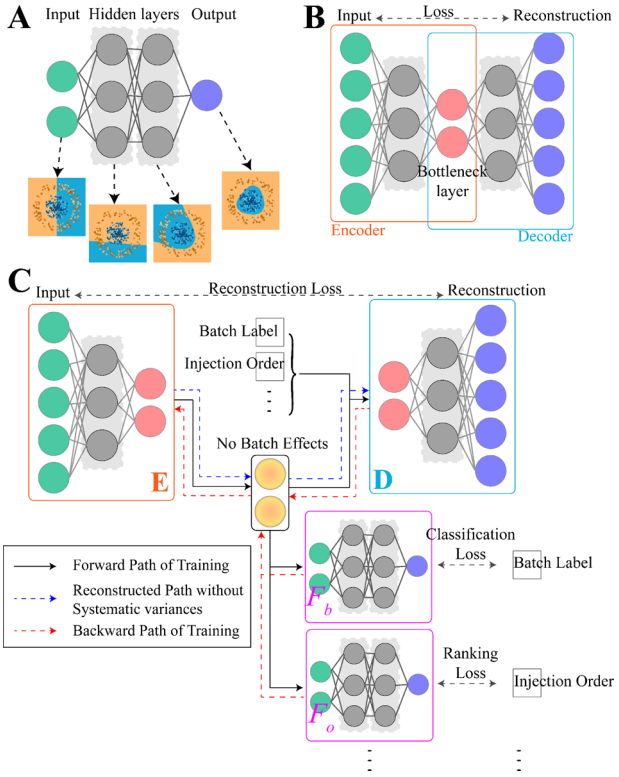

Figure 1. Schematic diagram of DNNs (A), AEs (B), and NormAE (C). For NormAE, the black solid path and red dotted line indicate the forward and backward calculation steps of the backpropagation algorithm, respectively. The blue dotted path indicates batch effect removal step after training.

intensities of peaks into latent representations that contain biological information without interbatch effects. Then decoder D achieves reconstruction using these representations to remove interbatch effects. To optimize the AE, an adversarial training procedure and an discriminator F_b that is trained to classify the batch labels based on the latent representations are required. E is trained so that the latent representations make F_b achieve poor classification. This means that E needs to drop interbatch information to get such latent representations. Simultaneously, decoder D merges the batch labels and latent representations to reconstruct the original intensities. It ensures that E does not lose other useful information, particularly biological variations; that is, E, D, and F_b play the following minimax game with loss function $V(D, E, F_b)$:

$$\min_{D,E} \max_{F_b} V(D, E. F_b) = \text{loss}_{\text{rec}}(x, D(E(x), y^b)) - \lambda^b \text{loss}_{\text{disc-b}}(F_b(E(x)), y^b), \quad (1)$$

where x and y^b are the original intensities of peaks and batch labels of minibatch samples, respectively. λ^b is a hyperparameter that controls the strength of the adversarial regularization. $\text{Loss}_{\text{rec}}(x, x')$ is the reconstruction loss of the AE. In this study, we chose MAE because it has sharper gradients than MSE:

$$\text{loss}_{rec}(x, x') = \sum_{x_i \in x, x_i' \in x'} \|x_i - x_i'\| \quad (2)$$

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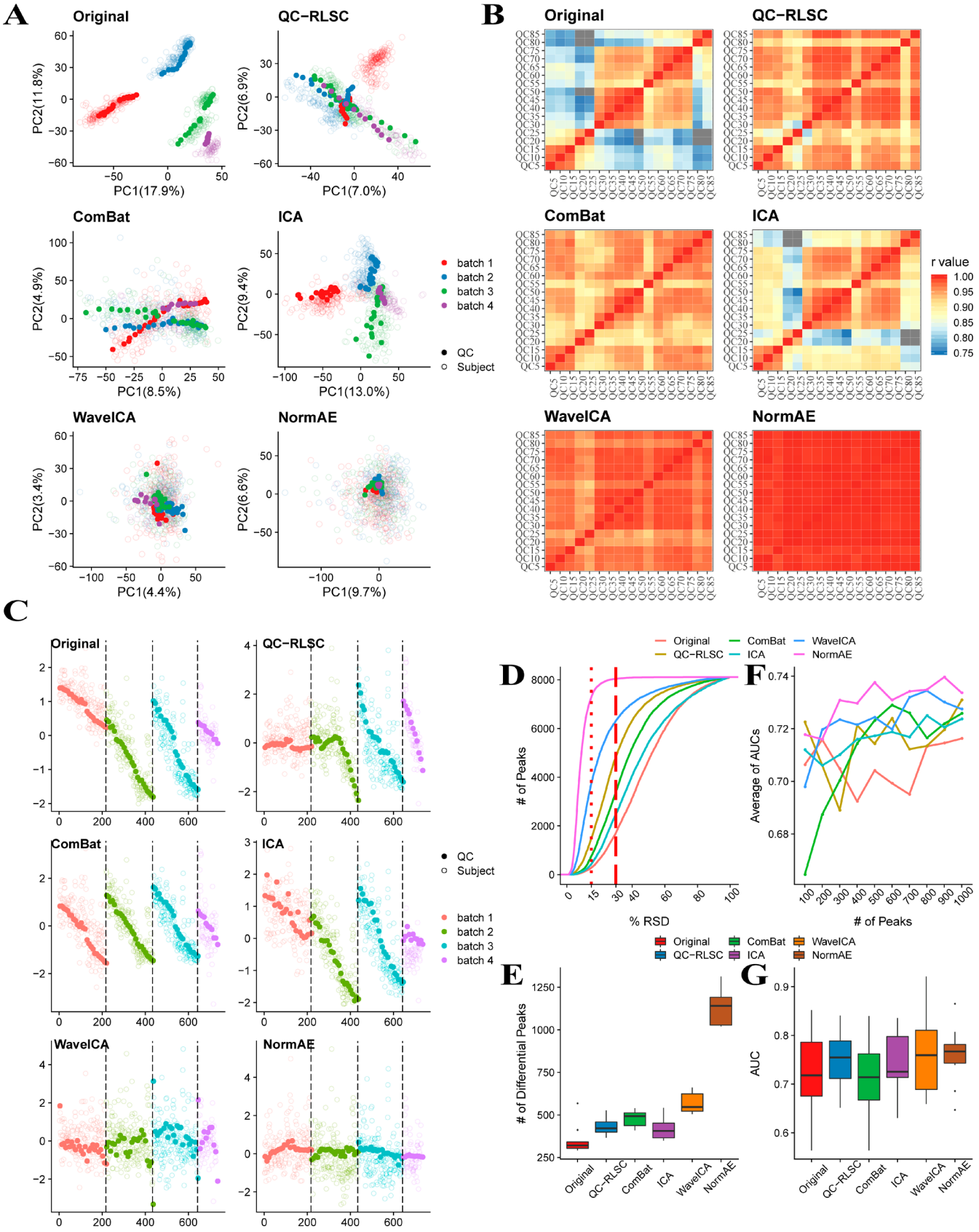

Figure 2. Results of the Amide data set: PCA score plots (A), heatmaps of the PCCs (B), intensity of peak M235T294 changing with the injection order (C), cumulative RSD curve of the QCs (D), number of differential peaks (E), average AUC values using the same number of peaks (F), and AUC values using selected peaks based on the feature selection pipeline (G) before and after applying each batch effect removal method. The four colored circles represent the different batches. The solid and open circles represent the QCs and subject samples, respectively.

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$\text{loss}_{\text{disc-b}}(p, y)$ is the loss function used to classify batch labels, which minimizes the negative log-likelihood:

$$\text{loss}_{\text{disc-b}}(c, y) = -\sum_{c_i \in c, y_i \in y} \log P(y_i | c_i) \tag{3}$$

In practice, we must implement the game using an iterative, numerical approach. Optimizing F_b to completion in the inner loop of training is computationally prohibitive, and on finite data sets would result in overfitting. Instead, we alternate between one step of optimizing F_b and one step of optimizing AE (E and D). This results in F_b being maintained near its optimal solution, provided E changes sufficiently slowly.[27,30]

For real metabolomics data, we may need to remove more batch effects than interbatch labels. For example, intrabatch effects cannot be characterized by batch labels but are related to injection orders. Therefore, we can add another discriminator F_o to sort injection orders, so there are two discriminators trained in the inner loop of training and two adversarial regularizations for training the AE:

$$\begin{aligned} &\min_{D,E} \max_{F_b, F_o} V(D, E, F_b, F_o) \\ &= \text{loss}_{\text{rec}}(x, D(E(x), y^b, y^o)) \\ &- \lambda^b \text{loss}_{\text{disc-b}}(F_b(E(x)), y^b) \\ &- \lambda^o \text{loss}_{\text{disc-o}}(F_o(E(x)), y^o, y^b), \end{aligned} \tag{4}$$

where λ^b and λ^o control the regularization strength from batch labels and injection orders. γ^o are the injection orders of minibatch samples. For metabolomics data, there are different intrabatch effects caused by instrument adjustment or column replacement between two batches. Therefore, samples are only sorted within a batch and we consider that interbatch effects have been identified and removed by F_b. We improved the loss of RankNet[31] to process the sorting task:

$$\begin{aligned} loss_{\text{disc-o}}(p, y^o, y^b) &= \sum_{(i,j) \in \mathcal{P}} \log(p_j) - \log(p_i) \\ \mathcal{P} &= \{(i, j) | (y_i^o > y_j^o) \wedge (y_i^b = y_j^b)\}, \end{aligned} \tag{5}$$

where i and j represent different samples. If there are more types of batch effects, then more discriminators can be added to remove these variations. The complete procedure is formally presented in Algorithm 1 in the Supporting Information.

Implementation. Deep learning models generally have to adjust most of the hyperparameters, such as the learning rate, number of hidden layers, and number of units in each hidden layer. We introduce the implementation of this study and believe that it will be applicable to most untargeted LC-MS metabolomics studies after a few adjustments.

The detailed network architecture is listed in Table S-1. We first pretrained the encoder E and decoder D with loss_{rec} for 1000 epochs, and then pretrained the discriminators F_b with $\text{loss}_{\text{disc-b}}$ and F_o with $\text{loss}_{\text{disc-o}}$ for 10 epochs. The iterative phase was trained for 700 epochs. All losses were minimized using the Adam optimizer with $\beta_1 = 0.5$, $\beta_2 = 0.9$. For pretraining E and D, the learning rate was 0.0002, whereas learning rates were 0.005 and 0.0005 for F_b and F_o, respectively. The minibatch size was 60. λ^b and λ^o were both 1. The NormAE model was implemented using PyTorch modules (version 1.2.0). The source code is available at https://github.com/luyiyun/NormAE.

RESULTS AND DISCUSSION

The well-known methods, ComBat, ICA, and QC-RLSC, were used in this study for comparison. They are representatives of location-scale, matrix factorization, and QCs-based methods, respectively. WaveICA was also selected because of its good performance for metabolomics data. Following the idea of Deng et al.,[19] the performance of the methods was mainly evaluated in two aspects: removal of batch effects and retention of biological information. The former was evaluated using the tightness of sample clustering in the PCA score plot, the distribution of the relative standard deviation (RSD) for all peaks in the QCs, and the trend plot of peak intensities as a function of the injection order. The average Pearson correlation coefficients (PCC) and average distances of QCs in the PCA score space were calculated as quantitative measurements. The latter was evaluated using the number of biomarkers discovered and predicted accuracy of CE versus CRC. Partial least-squares discriminant analysis (PLS-DA)[32] and the false discovery rate (FDR)[33] were used to identify biomarkers. Peaks of the FDR value less than 0.05, and variable importance in the projection (VIP) of PLS-DA greater than 1 in the training set were considered as potential biomarkers. Then RF[34] models were trained on the training set and tested on the test set using these differential peaks. The evaluation criterion for predictive accuracy was the area under receiver operating characteristic curve (AUC). Biomarker discovery and the evaluation of predictive accuracy were repeated using 10-fold cross-validation.

Amide Data Set. First, all five methods were used on the Amide mode data set. Figure 2A shows the PCA score plots of the original data set and normalized data sets for the five batch effect removal methods. ComBat, QC-RLSC, and ICA methods were not able to gather four batches together for either subject samples or QCs. The samples clustered together after using NormAE or WaveICA, but the performance of NormAE was better. The average distances of QCs in the PCA score plot are considered as the quantitative evaluation (Table 1). The NormAE method had the shortest average distances of QCs. This suggests that the NormAE method removed more batch effects than other methods.

Table 1. Average Distance of QCs in the PCA Score Plots (aDIST), Average PCCs between Pairs of QCs (aPCC), and Proportion of Peaks with RSDs (pRSD) within 15% and 30% of the QCs, Which Indicates That These Values Correspond to Each Method in Each Dataset

data set	method	aDIST	aPCC	pRSD15(%)	pRSD30(%)
amide	original	56.550	0.873	3.205	19.598
	NormAE	**7.383**	**0.997**	**89.794**	**99.100**
	WaveICA	18.968	0.972	49.279	76.211
	ComBat	36.394	0.941	9.244	40.897
	ICA	49.320	0.897	5.497	27.179
	QC-RLSC	23.361	0.944	17.416	56.785
T3	original	52.476	0.907	8.775	38.846
	NormAE	**14.075**	**0.997**	**91.872**	**99.121**
	WaveICA	27.417	0.957	50.793	73.041
	ComBat	34.284	0.926	25.262	56.200
	ICA	38.344	0.876	29.985	52.099
	QC-RLSC	28.093	0.938	34.110	64.962

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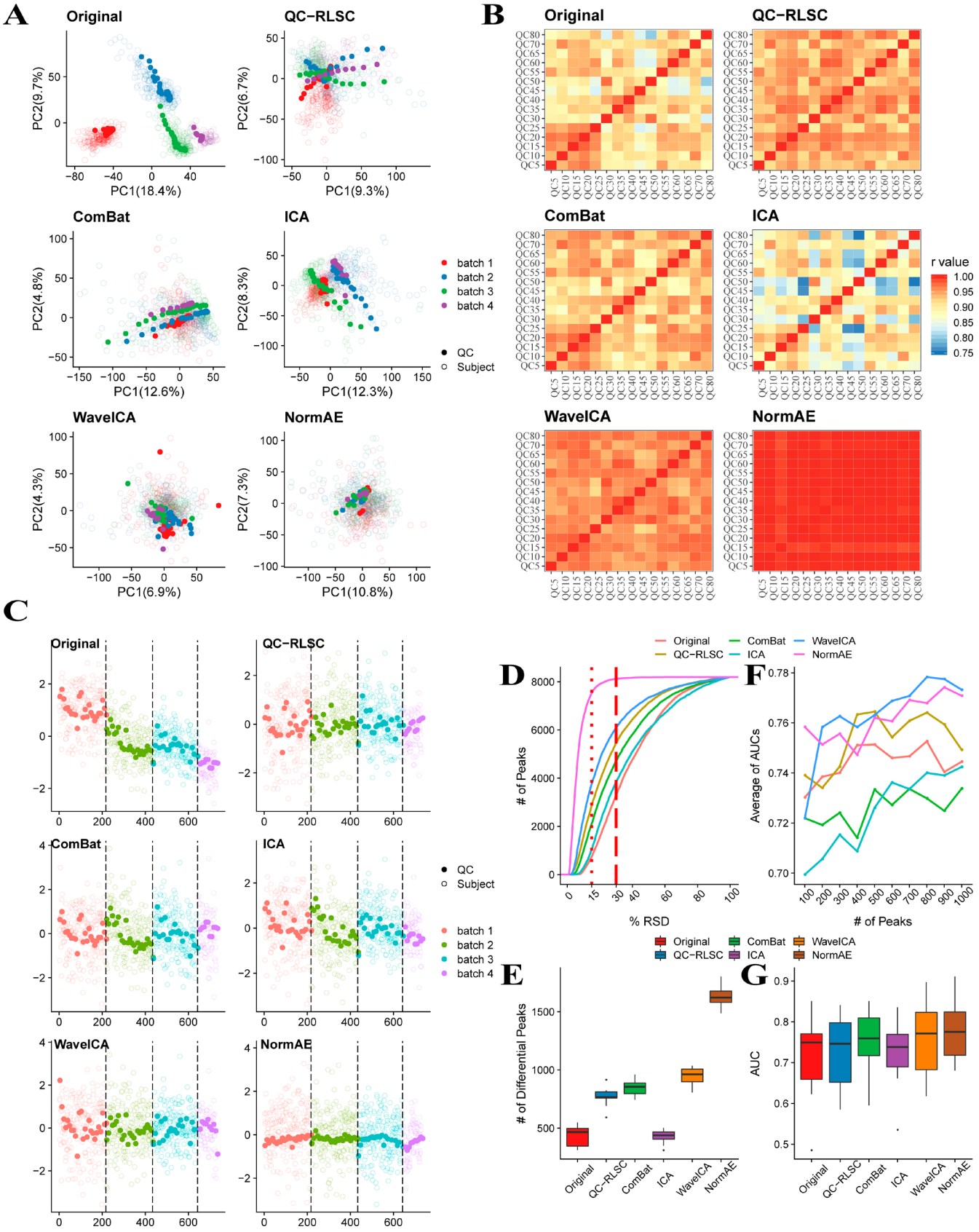

Figure 3. Results of the T3 data set: PCA score plots (A), heatmaps of the PCCs (B), intensity of peak M63T31 changing with the injection order (C), cumulative RSD curve of the QCs (D), number of differential peaks (E), average AUC values using the same number of peaks (F), and AUC values using selected peaks based on the feature selection pipeline (G) before and after applying each batch effect removal method. The four colored circles represent the different batches. The solid and open circles represent the QCs and subject samples, respectively.

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As mentioned above, QCs should have strong correlation relationships with each other. Therefore, PCCs among QCs can be used to evaluate the removal of batch effects. Table 1 shows that NormAE attained the highest values. Other methods had different levels of improvement, but were lower than NormAE. This can also be seen in the heatmap of PCCs of QCs (Figure 2B): the larger the area of the red region, the better the similarities between QCs. The heatmap of the original data shows that a large number of pairs of QCs had lower PCCs than 0.85, and several batches can be clearly observed. The improvement of the ICA method was very limited, whereas QC-RLSC and the ComBat method had similar performance that was better than that of ICA. WaveICA had good results, but a minor inconsistency was observed, which was that the PCCs of QC20−QC25 samples were lower than those of other pairs. The NormAE method made the PCCs of all QC pairs close to 1, and no batch phenomenon was observed.

The performance of each method given by the cumulative RSD curve of QCs is shown in Figure 2D. It indicates that NormAE reduced the variances caused by batch effects, and outperformed the other batch effect removal methods. After applying the NormAE method, the proportion of peaks that satisfied the FDA 15% criteria for measuring sample repeatability increased about 30 times and it was higher than the other methods (Table 1). We next performed the intensity drift over the injection order during LC/MS analysis using an example peak (M235T294: m/z 234.9, retention time 294.314s) (Figure 2C). The intensity of QCs normalized by NormAE was the most consistent compared with other methods, which may be the reason for the low RSD of NormAE. Although WaveICA also removed most of the batch effects, some slight fluctuations and a large variance of QCs were still observed. ComBat was unable to handle intrabatch effects. QC-RLSC had no ability to fit the discontinuity between two batches, so it worked on the first batch but failed on the others. The ICA method made a few improvements. There were other examples (Figures S-1−S-5) that all supported the above results.

Batch effects mask real biological variation, so batch effect removal methods should be able to expose more biological effects than the original data because of accurate normalization. The classification of CRC and CE was used to demonstrate the performance of exposing biological information. The feature selection pipeline was applied for each data set, and then medians of the number of selected peaks were calculated. As shown in Figure 2E, the median number of differential peaks in the original data was 322. After applying the methods, the medians were 1140, 546, 494, 421, and 406 for NormAE, WaveICA, ComBat, QC-RLSC, and ICA, respectively. Data normalized by NormAE had the most differential peaks, which was almost twice that of WaveICA. The test AUC values for the RF models built using these peaks are shown in Figure 2G. NormAE achieved the best performance. The results demonstrate that the NormAE method detected more potential metabolites than the other methods and improved the classification accuracy. To avoid the influence of peak numbers, the same number of peaks selected by PLS-VIP values for each method were used to estimate the RF models, and then averages of the test AUC were calculated (Figure 2F). The NormAE method performed best in terms of improving the classification accuracy for any number of peaks.

T3 Data Set. Next, we compared the five methods on the T3 mode data set. Similar to the Amide data set, the PCA score plot, and PCCs heatmap are shown to measure batch effect removal (Figure 3A). Samples distributed in strips indicate that the problem of intrabatch effects was unable to be solved by ComBat, ICA, and QC-RLSC. WaveICA gathered all samples together; however, the variation among QCs was still relatively large, which indicated the residues of batch effects. NormAE removed more batch effects nonlinearly, so the QCs were clustered more closely (Figure 3A). The heatmaps of PCCs are shown in Figure 3B. The original data set can be observed clearly for the four batches. QC_RLSC, ComBat, WaveICA, and NormAE all improved the similarity among QCs to different degrees, and NormAE was still the best. The average distance of QCs in the PCA score plot and average PCCs of QCs are shown in Table 1. Similar to the Amide data set, the NormAE method obtained the shortest average distance and highest average PCCs.

The cumulative RSD curve of the QCs for the T3 data set is shown in Figure 3D. The proportion of peaks with an RSD within 15% of the QCs was from 8.775 to 91.872 and that within 30% of the QCs was from 38.846 to 99.121 (Table 1). This indicates that NormAE removed more variance of the QCs than the other methods. An example peak (M63T31: m/z 63.0, retention time 31.273s) was chosen to show the intensity drift among the four batches (Figure 3C). Only NormAE reduced the high variation among OCs while normalizing multiple batches of samples to the same location. Therefore, NormAE had the best performance in the cumulative RSD curve. Other peaks (Figures S-6−S-10) supported the above results. The above results all indicate that NormAE performed well in removing batch effects.

The classification of CRC and CE was also implemented on the T3 data set. Figure 3E shows that the selected peaks based on data normalized by NormAE were larger than those of the other batch effect removal methods. Although the differences among all methods were not large, the RF model based on the NormAE method had the best test AUC value (Figure 3G). Then the same number of peaks selected by PLS-VIP values for each method were used to estimate the RF models. The performance of NormAE was second only to WaveICA, and the difference between the two was very small (Figure 3F).

Nonlinear Disentanglement. To prove that NormAE can disentangle and remove batch effects nonlinearly, the classifications of batch labels using nonlinear discriminator were implemented before and after batch effect removal. If batch effects are removed nonlinearly, then the test predicted accuracy based on normalized data should be lower than that on the original data. The RF model was selected as the discriminant model. The results show that only NormAE reduced the average accuracy from 1.000 to 0.855 and 0.847 for Amide and T3 data, respectively. Although the above results for WaveICA were relatively good and showed no differences among the four batches, the accuracies of batch label classification were still high (0.994 for Amide data and 0.999 for T3 data). There were still many batch effects hidden in the data calibrated by WaveICA, which may have had an impact on downstream analysis.

Comparison and Discussion. We found an interesting phenomenon from the above results: the better the performance of the method, the stronger the nonlinear fitting ability. ICA is the only method in the study that does not consider any nonlinear relationship. The ComBat model includes a

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multiplicative factor and is estimated by empirical prior estimation, so a partial nonlinear relationship can be detected. WaveICA improves the ICA method using a WT, which makes the nonlinear ability stronger than that of ComBat; therefore, WaveICA performs better than ICA and ComBat. DNNs are one of the most powerful models for nonlinear fitting, so NormAE removes most batch effects and achieves the best calibration.

Strengthening the nonlinearity also allows the batch effects to be separated from other effects cleanly. A benefit of this is that slight biological variations are retained as much as possible when removing batch effects. Other methods do not have a sufficient ability to decouple the combination of slight biological variations and batch effects. This may be the reason that there were more peaks and better AUC values based on data normalized by NormAE than by the other methods.

In theory, a good batch effect removal method for LC-MS data should remove both interbatch and intrabatch effects. However, only interbatch effects are taken into account by ComBat and ICA. By contrast, NormAE can integrate two adversaries, one for classifying batch labels (F_b) and the other for sorting injection orders (F_o), to remove all batch effects indirectly. It is not difficult to describe the versatility of the NormAE model. If there is other known batch information, such as temperature and pressure, the NormAE model can integrate it into the model similar to batch labels and injection orders.

QC-RLSC used nonlinear smoothing spline functions, so it should have a strong ability to discover nonlinear relationships. However, there was the problem of the scarcity of training QCs increasing the risk of overfitting. It can be observed in the PCA score plot that QCs were gathered together and subject samples were not (Figures 2A and 3A). Another problem is that discontinuity between two batches cannot be fitted by smooth functions, which was observed in Figures 2C and 3C. Discontinuities of intensity may originate from replacing the chromatography column and recalibrating instrument, and is very common in metabolomics research. NormAE provides additional ideas for solving these problems. There were as many samples as possible that could be used by NormAE because it indirectly removes batch effects using adversarial learning. Furthermore, we improved the loss function of RankNet[31] to respond to intrabatch effects exclusively, and interbatch effects were removed by $loss_{disc-b}$ using batch labels. These two effects were processed separately, so NormAE was not sensitive to the existence of discontinuities (Figures 2C and 3C).

The disadvantages of NormAE are the failure on a small sample size and sensitivity to outliers. Generally, these are common problems encountered by DNNs. In future work, we will improve the current algorithm to solve these problems. A limitation of this study is that the selected peaks are unable to be further analyzed and identified; therefore, it is difficult to determine whether NormAE truly screens more meaningful metabolites. In future work, we will identify some of the peaks selected using NormAE to determine whether batch effect removal methods can reveal real biological information.

■ CONCLUSIONS

For large-scale untargeted metabolomics studies, the influence of various batch effects is inevitable and difficult to remove. We proposed a novel method called NormAE to remove batch effects for large-scale LC-MS untargeted metabolomics data. The NormAE method performed well, not only for removing batch effects but also in revealing more meaningful potential metabolites. Using two metabolomics data sets based on an LC-MS system, the NormAE method outperformed other methods in our study.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.9b05460.

Detailed model architectures; accuracy of batch label classification; pseudo code for the complete algorithm; plots of intensity drift over injection orders of the other example peaks; tSNE score plots of each method on each data set; and PCA score plots for different sample sizes and log transformation (PDF)

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All authors have approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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